Practice workbook · Identify ear parts and use the sound path to explain hearing and hearing loss

Naming ear structures, roles of pinna, eardrum, ossicles, cochlea, Eustachian tube and semicircular canals, and how glue ear, pressure, ageing or damage reduce hearing.

10 question parts: 1 worked-example retry and 9 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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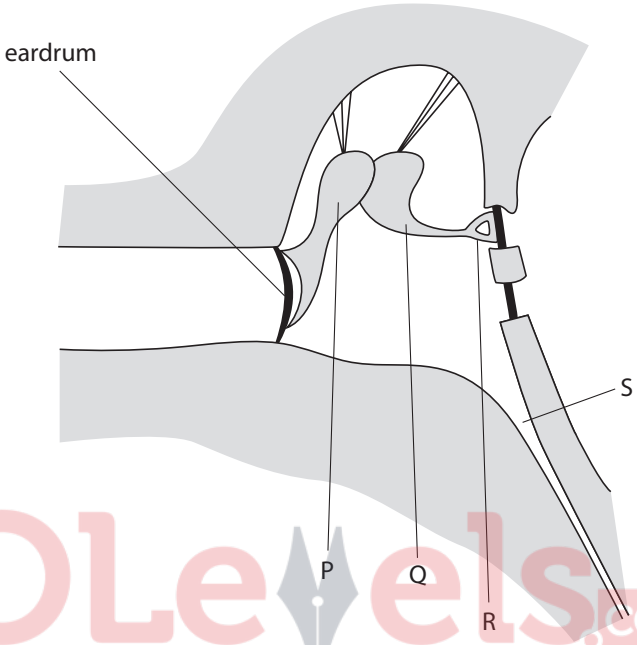
Type ID: HBI-EAR-PARTS-AND-HEARING-LOSS

1. Question 5 · part b.i · 4 marks

November 2020 Paper 2

Worked-example retry · Attempt part b.i only.

5 The diagram shows part of a human ear.



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(a) (i) Give the name of structure S.

(1)

.....

(ii) Name three parts of the ear that are **not** shown on the diagram.

(3)

1 .....

2 .....

3 .....

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1. Question 5 · part b.i · 4 marks

November 2020 Paper 2

Worked-example retry · Attempt part b.i only.

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(b) (i) Describe the functions of structures P, Q and R.

(4)

(ii) What is the name of the medium directly surrounding structures P, Q and R?

(1)

- ☒ **A** air
- ☒ **B** bone
- ☒ **C** fluid
- ☒ **D** oxygen

(c) Explain the importance of structure S in the process of hearing.

(3)

(Total for Question 5 = 12 marks)

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1. Question 5 · part b.i · 4 marks · Working and answer

Working and answer

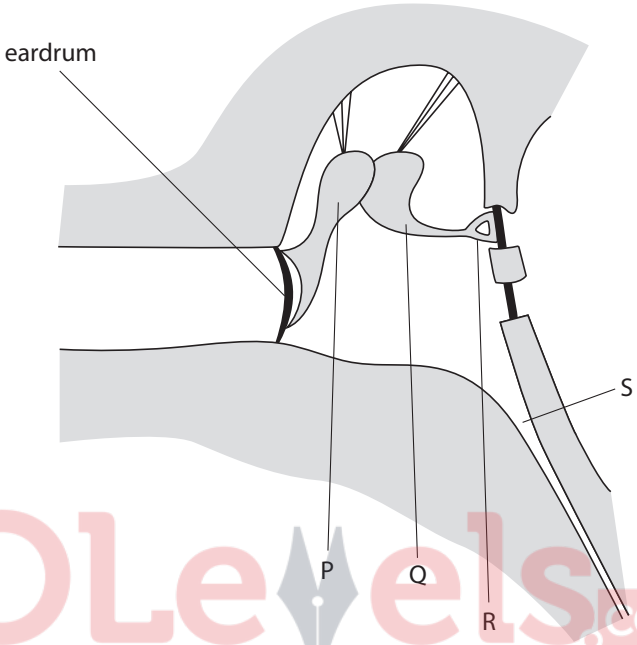
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2. Question 5 · part a.i · 1 mark

November 2020 Paper 2

Further practice · Attempt part a.i only.

5 The diagram shows part of a human ear.



(a) (i) Give the name of structure S. **(1)**

.....

(ii) Name three parts of the ear that are **not** shown on the diagram. **(3)**

1 .....

2 .....

3 .....

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2. Question 5 · part a.i · 1 mark · Working and answer

Working and answer

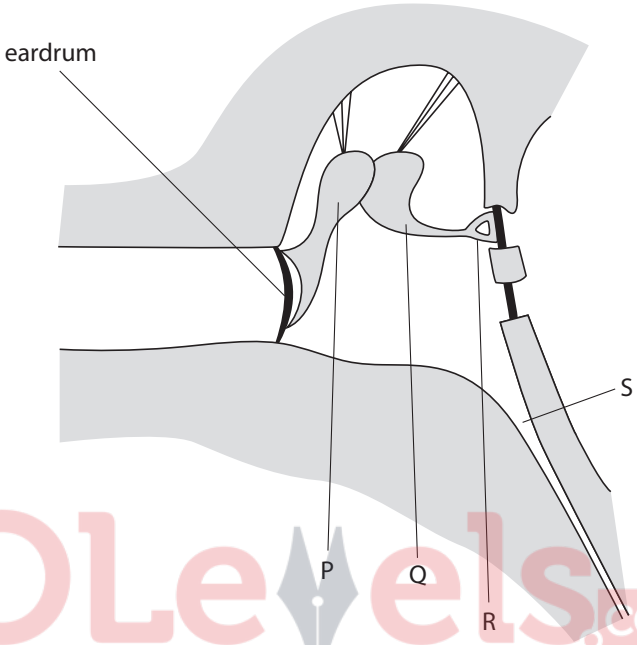
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3. Question 5 · part a.ii · 3 marks

November 2020 Paper 2

Further practice · Attempt part a.ii only.

5 The diagram shows part of a human ear.



(a) (i) Give the name of structure S.

(1)

.....

(ii) Name three parts of the ear that are **not** shown on the diagram.

(3)

1 .....

2 .....

3 .....

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3. Question 5 · part a.ii · 3 marks

November 2020 Paper 2

Further practice · Attempt part a.ii only.

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(b) (i) Describe the functions of structures P, Q and R.

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- ☒ **A** air
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- ☒ **D** oxygen

(c) Explain the importance of structure S in the process of hearing.

(3)

(Total for Question 5 = 12 marks)

3. Question 5 · part a.ii · 3 marks · Working and answer

Working and answer

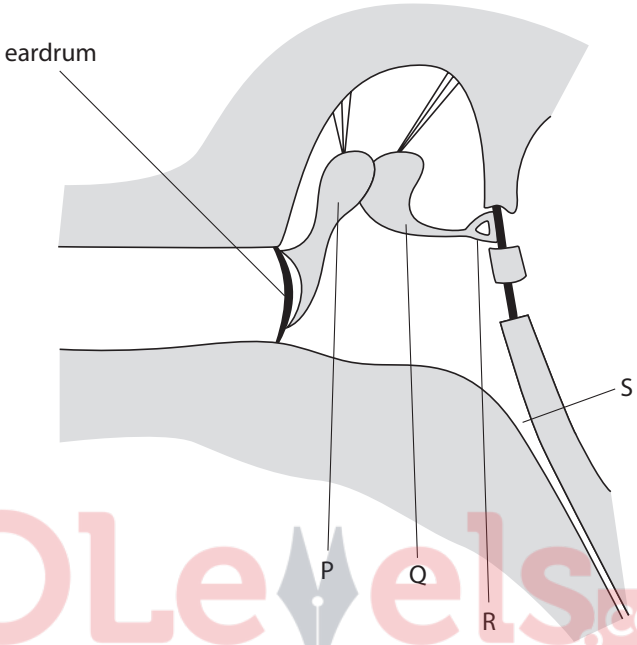
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4. Question 5 · part b.ii · 1 mark

November 2020 Paper 2

Further practice · Attempt part b.ii only.

5 The diagram shows part of a human ear.



(a) (i) Give the name of structure S.

(1)

.....

(ii) Name three parts of the ear that are **not** shown on the diagram.

(3)

1 .....

2 .....

3 .....

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4. Question 5 · part b.ii · 1 mark

November 2020 Paper 2

Further practice · Attempt part b.ii only.

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(b) (i) Describe the functions of structures P, Q and R.

(4)

(ii) What is the name of the medium directly surrounding structures P, Q and R?

(1)

- ☒ **A** air
- ☒ **B** bone
- ☒ **C** fluid
- ☒ **D** oxygen

(c) Explain the importance of structure S in the process of hearing.

(3)

(Total for Question 5 = 12 marks)

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4. Question 5 · part b.ii · 1 mark · Working and answer

Working and answer

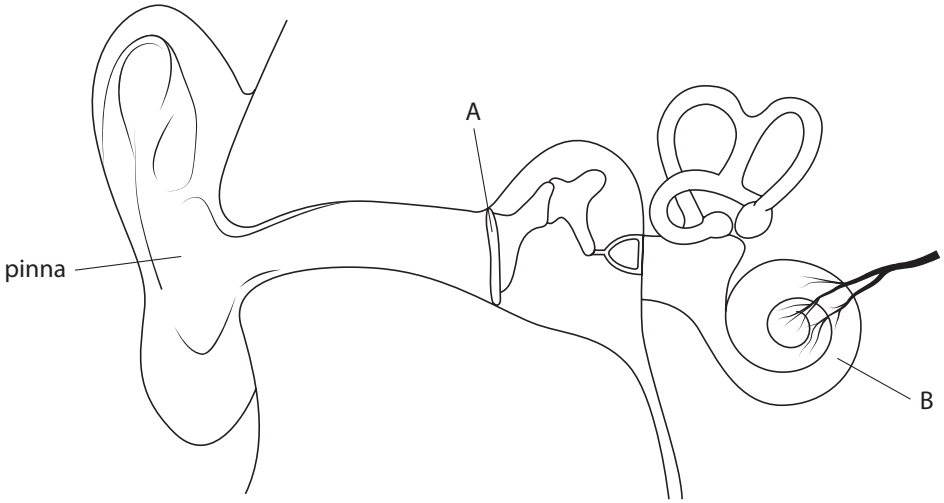
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5. Question 5 · part a.i · 2 marks

June 2022 Paper 2

Further practice · Attempt part a.i only.

5 The diagram shows a section through the human ear.



(a) (i) Name parts A and B. **(2)**

A .....

B .....

(ii) Describe the function of the pinna. **(2)**

.....

.....

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.....

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5. Question 5 · part a.i · 2 marks · Working and answer

Working and answer

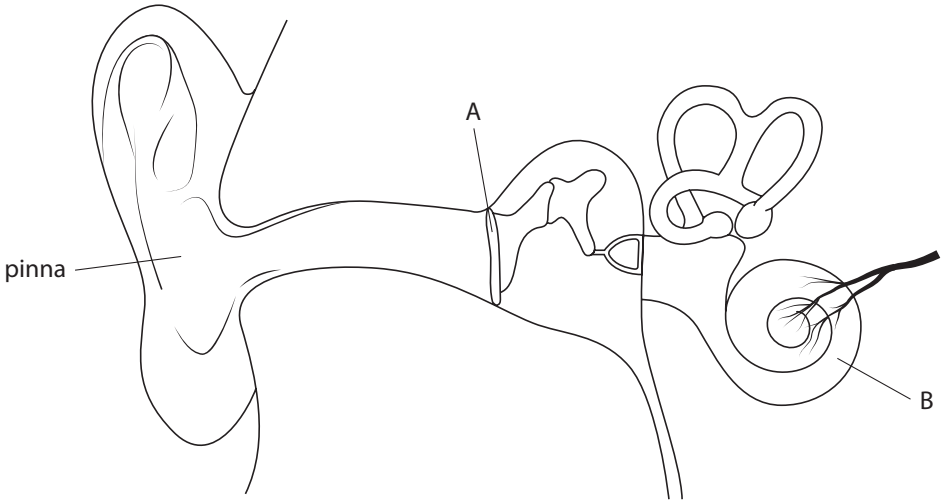
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6. Question 5 · part a.ii · 2 marks

June 2022 Paper 2

Further practice · Attempt part a.ii only.

5 The diagram shows a section through the human ear.



(a) (i) Name parts A and B. **(2)**

A .....

B .....

(ii) Describe the function of the pinna. **(2)**

.....

.....

.....

.....

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6. Question 5 · part a.ii · 2 marks

June 2022 Paper 2

Further practice · Attempt part a.ii only.

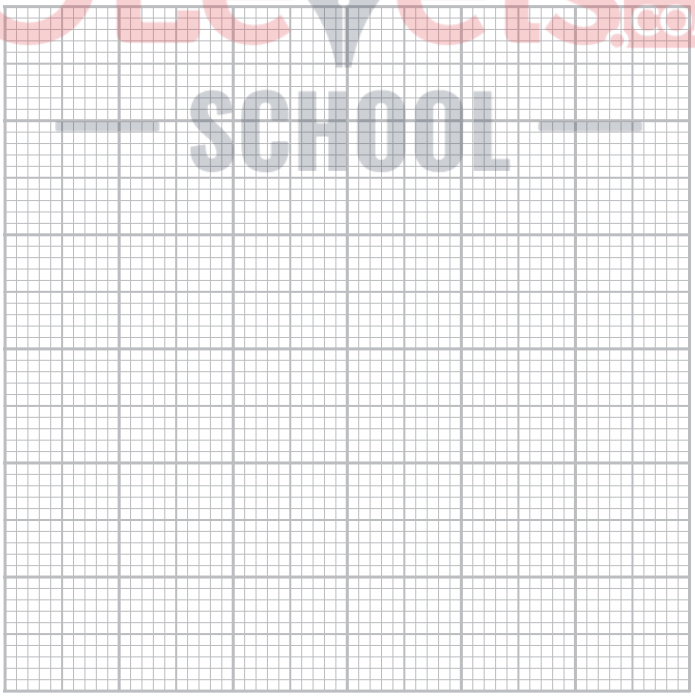
(b) The range of frequencies audible to the human ear can vary with age.

The table shows the maximum frequency detected by people of different ages.

Age in years	Maximum frequency detected in kHz
20	20
30	18
40	16
50	14

(i) Draw a bar chart to show how the maximum frequency detected varies with age.

(4)



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6. Question 5 · part a.ii · 2 marks · Working and answer

Working and answer

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7. Question 5 · part b.iii · 3 marks

January 2023 Paper 1

Further practice · Attempt part b.iii only.

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5 (a) A student reads that as people get older their ability to hear high frequency sounds decreases.

(i) Describe an investigation a student could do to show if this is correct.

(3)

(ii) Give a reason why hearing loss occurs as people get older.

(1)

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7. Question 5 · part b.iii · 3 marks

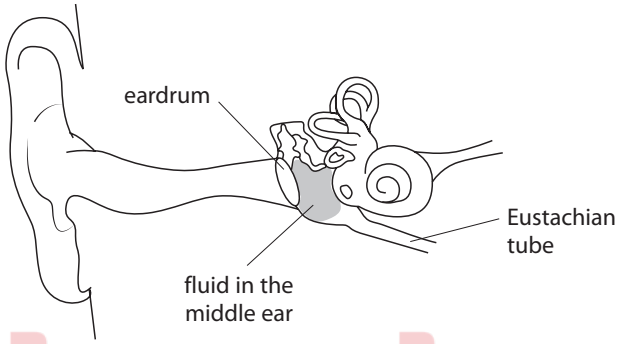
January 2023 Paper 1

Further practice · Attempt part b.iii only.

(b) Glue ear is a condition that can affect children and causes hearing loss.

Fluid accumulates in the middle ear, as shown in the diagram.

As the child grows, the slope of the Eustachian tube increases and the fluid drains.



(i) Explain how the fluid in the middle ear causes hearing loss.

(3)

(ii) Explain why children with glue ear get more bacterial ear infections than other children.

(2)

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7. Question 5 · part b.iii · 3 marks

January 2023 Paper 1

Further practice · Attempt part b.iii only.

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(iii) Describe the role of the ear in maintaining a person's balance.

(3)

(Total for Question 5 = 12 marks)

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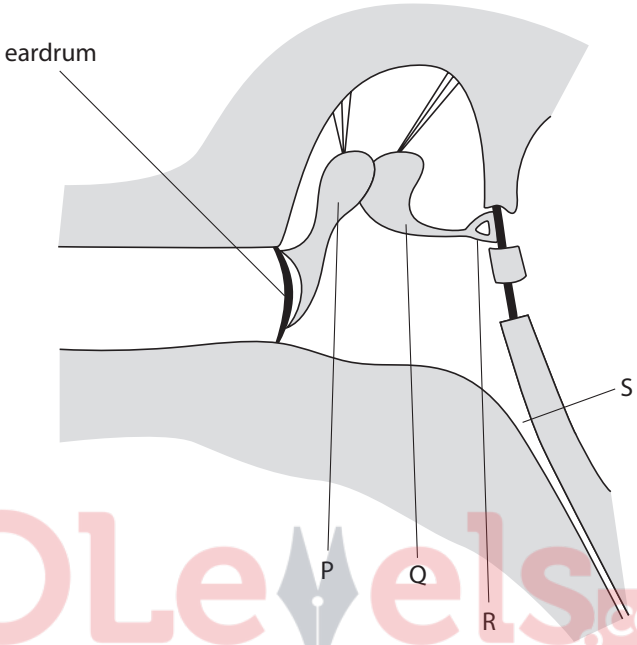
Working and answer

8. Question 5 · part c · 3 marks

November 2020 Paper 2

Further practice · Attempt part c only.

5 The diagram shows part of a human ear.



(a) (i) Give the name of structure S.

(1)

.....

(ii) Name three parts of the ear that are **not** shown on the diagram.

(3)

1 .....

2 .....

3 .....

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8. Question 5 · part c · 3 marks

November 2020 Paper 2

Further practice · Attempt part c only.

(b) (i) Describe the functions of structures P, Q and R.

(4)

(ii) What is the name of the medium directly surrounding structures P, Q and R?

(1)

- ☒ **A** air
- ☒ **B** bone
- ☒ **C** fluid
- ☒ **D** oxygen

(c) Explain the importance of structure S in the process of hearing.

(3)

(Total for Question 5 = 12 marks)

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8. Question 5 · part c · 3 marks · Working and answer

Working and answer

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9. Question 5 · part b.i · 3 marks

January 2023 Paper 1

Further practice · Attempt part b.i only.

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5 (a) A student reads that as people get older their ability to hear high frequency sounds decreases.

(i) Describe an investigation a student could do to show if this is correct.

(3)

(ii) Give a reason why hearing loss occurs as people get older.

(1)

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9. Question 5 · part b.i · 3 marks

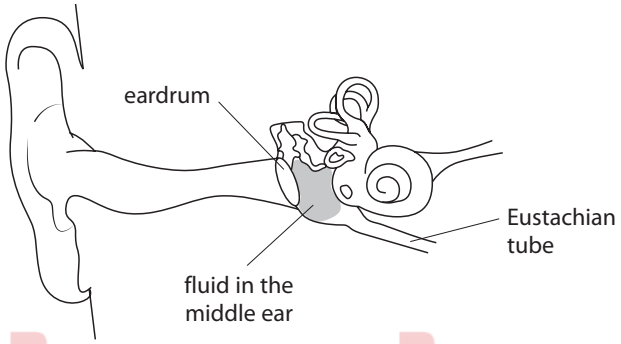
January 2023 Paper 1

Further practice · Attempt part b.i only.

(b) Glue ear is a condition that can affect children and causes hearing loss.

Fluid accumulates in the middle ear, as shown in the diagram.

As the child grows, the slope of the Eustachian tube increases and the fluid drains.



(i) Explain how the fluid in the middle ear causes hearing loss.

(3)

(ii) Explain why children with glue ear get more bacterial ear infections than other children.

(2)

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P 7 2 2 7 1 A 0 1 4 2 4

9. Question 5 · part b.i · 3 marks · Working and answer

Working and answer

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10. Question 5 · part a.ii · 1 mark

January 2023 Paper 1

Further practice · Attempt part a.ii only.

5 (a) A student reads that as people get older their ability to hear high frequency sounds decreases.

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(3)

(ii) Give a reason why hearing loss occurs as people get older.

(1)

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10. Question 5 · part a.ii · 1 mark

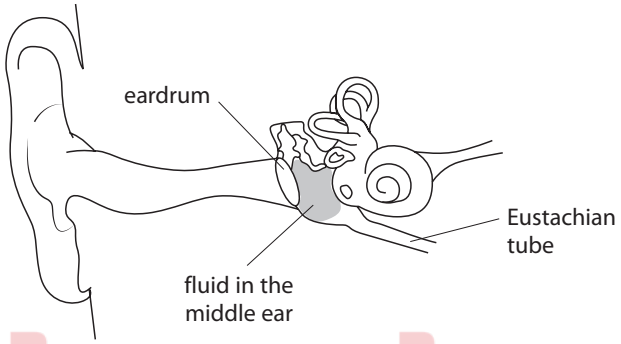
January 2023 Paper 1

Further practice · Attempt part a.ii only.

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(i) Explain how the fluid in the middle ear causes hearing loss.

(3)

(ii) Explain why children with glue ear get more bacterial ear infections than other children.

(2)

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P 7 2 2 7 1 A 0 1 4 2 4

10. Question 5 · part a.ii · 1 mark · Working and answer

Working and answer

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